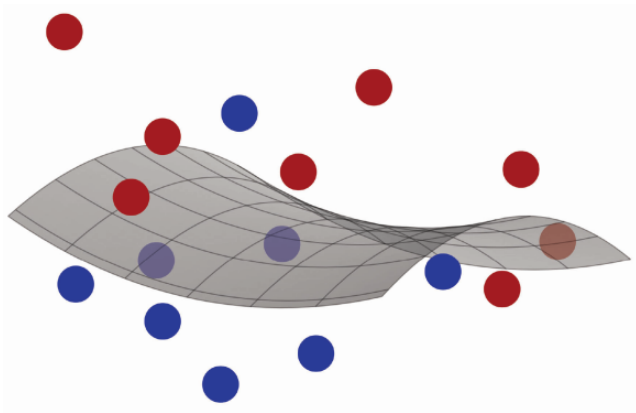


Rademacher Complexity and VC-dimension



Foundations of Machine Learning

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Context and Motivation

Setup: Distribution $\mathcal{D}$ over $(x, y) \in \mathcal{X} \times \mathcal{Y}$, loss ℓ , hypothesis class $\mathcal{H}$.

- True risk: $L_{\mathcal{D}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(h(x), y)]$,
- Empirical risk: $L_S(h) = \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i)$ for $S \sim \mathcal{D}^m$.

Agnostic Probably Approximately Correct (PAC) learnability

A class is *agnostically PAC learnable* if, with

$$m \geq m_{\mathcal{H}}(\epsilon, \delta) \text{ samples,}$$

with probability $\geq 1 - \delta$ over the choice of $S \sim \mathcal{D}^m$.

$$L_{\mathcal{D}}(\text{ERM}_{\mathcal{H}}(S)) \leq \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon$$

Generalisation Gap

$$\begin{aligned}\mathcal{E}(h) &:= L_{\mathcal{D}}(h) - L_S(h) \\ &= \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(h(x), y)] - \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i)\end{aligned}$$

- We want to learn a model h^* that minimizes the True risk $L_{\mathcal{D}}(h^*)$
- But we don't have access to $\mathcal{D}$, only to $S \sim \mathcal{D}^m$
- So we minimize $L_S(h) = \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i)$ (ERM) and learn $h^* = \text{ERM}_{\mathcal{H}}(S)$
- But we still need to know if the Generalisation gap is small enough:

If large $\Rightarrow$ overfitting

Rademacher complexity

Rademacher complexity captures the richness of a hypothesis class $\mathcal{H}$ by measuring how well it can fit random noise.

Empirical Rademacher complexity

With $\mathcal{G} = \{g : (x, y) \mapsto \ell(h(x), y) \mid h \in \mathcal{H}\}$
and $\forall S \sim \mathcal{D}^m, \forall g \in \mathcal{G}, g(S) = (g(x_i, y_i))_{1 \leq i \leq m}$

$$\hat{\mathfrak{R}}_S(\mathcal{G}) := \mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \frac{\sigma \cdot g(S)}{m} \right]$$

Where $\sigma = (\sigma_1, \dots, \sigma_m)$, with $\sigma_i \in \{-1, 1\}$ such that $\sigma_i \sim \mathfrak{B}(1/2)$

Rademacher complexity

$$\mathfrak{R}_m(\mathcal{G}) := \mathbb{E}_{S \sim \mathcal{D}} [\hat{\mathfrak{R}}_S(\mathcal{G})]$$

First generalisation bound

Theorem

$\forall \epsilon > 0$, with probability $1 - \delta$ over the draw of $S \sim \mathcal{D}^m$

$$L_{\mathcal{D}}(h) - L_S(h) \leq 2 \mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(1/\delta)}{2m}}$$

$$L_{\mathcal{D}}(h) - L_S(h) \leq 2 \hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}$$

- Computing Rademacher complexity can be difficult.
- This measure is data-dependant (it depends on the underlying distribution $\mathcal{D}$)

Growth function

We can bound the Rademacher complexity using:

Growth function

$\forall m \geq 1$

$$\Pi_{\mathcal{H}}(m) := \max_{S \sim \mathcal{D}^m} \# \{ (h(x_1), \dots, h(x_m)) \mid h \in \mathcal{H} \}$$

- It is the maximum number of distinct ways to classify m points using hypotheses space $\mathcal{H}$ estimators/models.
- This is another measure of richness but doesn't depend on the data distribution $\mathcal{D}$

Using some result called *Massart lemma* we can derive:

Corollary

$\forall m \geq 1$

$$\hat{\mathfrak{R}}_m(\mathcal{G}) \leq \sqrt{\frac{2 \log(\Pi_{\mathcal{H}}(m))}{m}}$$
$$\Rightarrow L_{\mathcal{D}}(h) - L_S(h) \leq \sqrt{\frac{2 \log(\Pi_{\mathcal{H}}(m))}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}}$$

- We still need to compute $\Pi_{\mathcal{H}}(m)$, $\forall m \geq 1$.

VC dimension

- For m points, the maximum number of ways they can be classified is 2^m
- So $\Pi_{\mathcal{H}}(m) \leq 2^m$ and $\Pi_{\mathcal{H}}(m) = 2^m$
 $\Rightarrow \exists S$ a set of points for which $\mathcal{H}$ can label the points in every possible way.

In this case we say that S is *shattered* by $\mathcal{H}$.

for example, with $\mathcal{H}$ the linear classifiers:



VC dimension

The VC dimension is the size of the largest set that can be shattered by $\mathcal{H}$.

$$\text{VCdim}(\mathcal{H}) := \max \{m \mid \Pi_{\mathcal{H}}(m) = 2^m\}.$$

example: linear classifiers in $\mathcal{X} \subset \mathbb{R}^d$ for $d = 2$:

We previously saw that $\Pi_{\mathcal{H}}(3) = 2^3$
 $\Rightarrow \text{VCdim}(\mathcal{H}) \geq 3$



VC dimension

- More generally in $\mathbb{R}^d$ for linear classifiers, $\text{VCdim}(\mathcal{H}) = 2d + 1$

Theorem

If $\text{VCdim}(\mathcal{H}) = d'$, $\forall m \geq d'$

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d'}\right)^{d'} = O\left(m^{d'}\right)$$

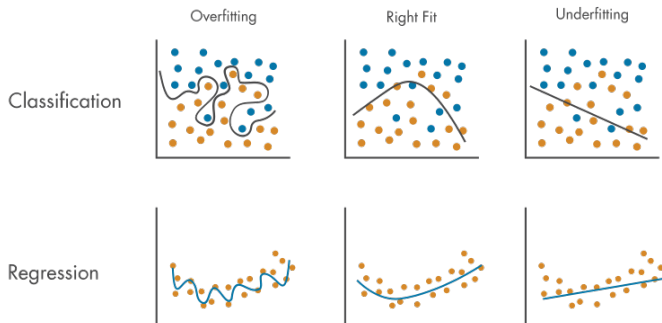
$\Rightarrow \forall \delta > 0$, with probability $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$

$$\begin{aligned} L_{\mathcal{D}}(h) - L_S(h) &\leq \sqrt{\frac{2d' \log(2m/d')}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} \\ &= O\left(\sqrt{\frac{\log(m/d')}{m/d'}}\right) \end{aligned}$$

- If the number of samples grows, the Generalisation Gap decreases
- If the VC dimension d' grows, the Generalisation Gap increases

VC dimension & overfitting

- If the VC dimension is too large wrt the number of data, the Hypothesis class is "too rich", which leads to overfitting.



- Knowing the VC dimension of a Hypothesis class can help us determine a lower bound on the number of needed data to not overfit.

Agnostic Probably Approximately Correct (PAC) learnability

A class is *agnostically PAC learnable* if, with

$$m \geq m_{\mathcal{H}}(\epsilon, \delta) \text{ samples,}$$

with probability $\geq 1 - \delta$ over the choice of $S \sim \mathcal{D}^m$.

$$L_{\mathcal{D}}(\text{ERM}_{\mathcal{H}}(S)) \leq \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon$$

- We say that a sample is $\frac{\epsilon}{2}$ -representative if $\sup_{h \in \mathcal{H}} L_{\mathcal{D}}(h) - L_S(h) < \frac{\epsilon}{2}$
- This is a sufficient condition to ensure PAC learnability.
- This condition is satisfied when $\sqrt{\frac{2d' \log(2m/d')}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} < \frac{\epsilon}{2}$.

VC dimension Lower bound on VC dimension

Theorem

If $\text{VCdim}(\mathcal{H}) = d' > 1$, then for all $m \geq 1$, there exists a learning algorithm $\mathcal{A}$, a distribution $\mathcal{D}$ over $\mathcal{X}$, and a target function $f \in \mathcal{H}$ such that

$$\mathbb{P}_{S \sim \mathcal{D}^m} \left[L_{\mathcal{D}}(h) > \frac{d'-1}{32m} \right] \geq \frac{1}{100}.$$

- If the hypothesis class has VC dimension d' , any learner may suffer a generalisation error at least proportional to $\frac{d'}{m}$
- This establishes a lower bound: no algorithm can guarantee PAC learnability with fewer than $O(d'/\epsilon^2)$ samples.

Questions

Thank you for your attention.

